

United States Federal Spending in Georgia

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I. INTRODUCTION

In a time where U.S. citizens desire to know more than ever where their tax dollars are being used, answering these questions is an important task. It is vital to provide clarity and rationale for the average tax payer that their money is going to secure areas that benefit and strengthen their community. Unfortunately in the past few years, waves of questionable and fraudulent programs benefiting off of government subsidy have been brought to light by several independent journalists, causing citizens concerns about what their tax dollars are being used for to increase drastically.

The goal of this project is to examine federal transactions addressed to the state of Georgia to discover hidden patterns within them. Specifically, it aims to answer three primary discovery questions about U.S. Federal Spending:

- What combinations of awarding agencies, recipients, and award types frequently occur in different sized regions within Georgia?
- "What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?"
- "What characteristics define anomalously high value transactions?"

Examining the state of Georgia in this manner should either provide reassurance to its citizens that their taxes are still primarily going to good places, or will expose an underlying problem within the state in need of correction.

The data set used in this study comes from "USASPENDING.gov", a website that collects and holds records of awards given to each state since 2008 [1]. It breaks down awards into multiple categories, including contracts, contract IDVs, grants, loans, direct payments, and other award types (Figure 1). Each award has several pieces of data giving context to the award, including the recipient name, awarding agency, amount granted, etc.

This study specifically uses the web page that contains records of all federal awards granted to Georgia. This study only takes into account awards in 2025 totaling over 20,000,000 dollars, as this is the most relevant area for considering current spending. The primary attributes of each data point that will be examined are the recipient name, the awarding agency, the award type, the award subtype, the award description and the obligated amount (Table 1). The obligated value of the award was chosen over the outlay amount due to a significant amount of missing outlays present in the dataset.

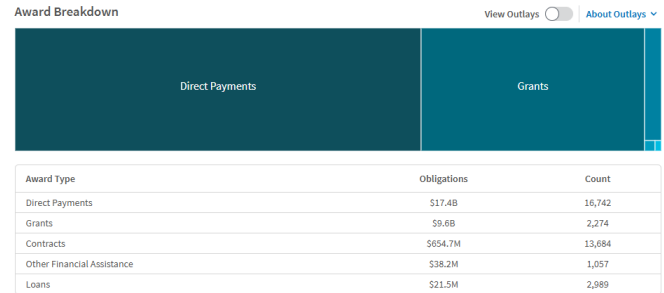


Fig. 1. A graph from "USASPENDING.gov" detailing the sum of all transaction types made to the state of Georgia from fiscal year 2008 to 2026.

transactions	
transaction_id	string
recipient	string
awarding_agency	string
award_type	string
award_subtype	string
description	string
obligations	float

TABLE I
DATA SCHEMA DESCRIBING THE BREAKDOWN OF EACH TRANSACTION AND ITS RELEVANT DATA. THIS STUDY PRIMARILY DEALS IN STRING VALUES, WITH THE EXCEPTION OF OBLIGATIONS, WHICH ARE RECORDED AS A FLOAT VALUE.

While the obligated amount may reduce clarity on where federal money is going, it will still provide context into the interests of the federal government and their spending habits. There may also be a quality issue in the description of each award, as some are more clear than others, while some barely provide a description at all. For this reason, the description is not as considered for analysis as the rest of the data, however it is still provided to give context to each award. After sectioning out the most important portion of time, and carefully considering which pieces of data are most relevant to the study, the data set used contains 1,478 rows of entries, each with 6 columns of data.

II. METHOD

Before preprocessing occurred, a box and whisker plot was generated to examine the number of outlier transaction values present in the set. The graph revealed that the top 25 percent

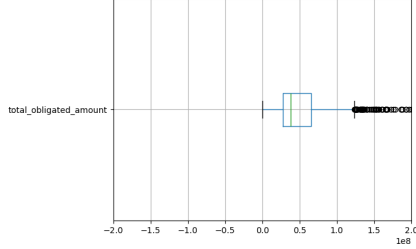


Fig. 2. Screenshot of the generated box and whisker plot for the total obligations.

of transactions were flagged as outlier values. The chances of the top fourth of the data being anomaly transactions was slim, so we needed to determine whether or not there were any anomaly transactions. To do this, we performed the Isolation Forest early, with the additional purpose of discovering the characteristics that make a high value transaction. We used the total obligations, total COVID supplements, and total IJJA supplements as parameters, with which we ran through the Isolation Forest. Once we found the actual number of anomalies, we examined the characteristics of each transaction and determined that none of the transactions were out of the ordinary transactions in their purpose, and therefore were all kept in the dataset.

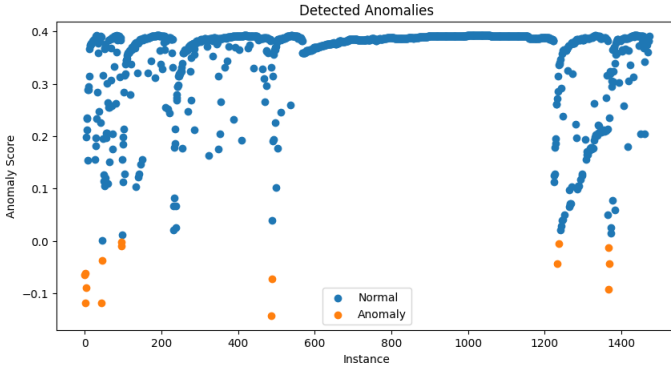


Fig. 3. Graph displaying every transaction marked as either normal or anomaly.

After we used an isolation forest to answer the third discovery question and flag potential outliers, we applied the apriori method of the association rule to answer the first discovery question. Because each transaction is made up of groups of varying qualitative characteristics, we believe apriori was the best choice to reveal common patterns within them. For this, we ran apriori four times under different sets of parameters:

- 1. Finding common sets of counties, county sizes, agencies, and transaction types.

- 2. Finding reoccurring patterns in combinations of agencies, business types, and counties.
- 3. Finding common combinations of words in transaction descriptions.
- 4. Finding interesting relationships between agencies and recipients.

While every run of apriori uses different parameters under different contexts, they all serve to answer the primary discovery questions. Each run treats each transaction like a set of parameters. The first run had the largest amount of parameters, those being the awarding agency, recipient, county, county size, and transaction type. The goal is to determine what purpose each regional size receives the most funding for, and which regions get funded the most. This approach used a support value of 0.01 and a minimum confidence of 0.7, which is done to find non-trivial patterns in a large dataset of multiple varying factors.

The second run uses the agency, business type, and county as parameters. Here, the aim is to find out which business types are funded by which agency, as well as which counties each business type can be found in. Just like the first set, the approach uses a support value of 0.01 and a minimum confidence of 0.7 once again to find non-trivial patterns in a large dataset of multiple varying factors.

The third run takes the description of each transaction and turns them into a list of words, each word being an item added to a set that represents one transaction. Here, we wanted to see if there were any common sets of words that would reveal any patterns of funding purposes common to Georgia. We continued to use apriori as opposed to text mining on this run to keep consistency and to directly compare the results of this run to the other three. We used a support value of 0.06 and a confidence value of 0.7, which allows the most amount of non-trivial words to appear in the table.

The final run only used two parameters, the agency and the recipient. The goal here is to find patterns in agency-recipient relationships. We want to find exclusive relationships, and relationships between agencies and multiple recipients. We used a very low support value of 0.006 and minimum confidence of 0.1, which allows us to see relations that only appear once, which is vital information to analyze.

Second, k-means clustering is used to answer the remaining discovery question by finding natural groupings of transaction recipients based on obligations and the number of contracts they receive. We chose k-means over DBSCAN because our dataset is large and uses large transactional values. In addition, there are hundreds of individual recipients to cluster, and k-means would perform this faster than DBSCAN. To do this, each recipient becomes represented by a coordinate based on the two factors (obligations on the x axis and number of awards on the y axis) and placed in a data frame. Based on the elbow method, we started at 3 clusters, but increased this to 5 to further distinguish members of the largest cluster apart. When running the k-means method, we ran it with 10 random initial clusters to increase our confidence that the groups presented by the clusters are natural.

III. RESULT (AGENCY - RECIPIENT - REGION RELATIONS)

After running apriori four times, we generated mostly trivial results that confirm known geographical patterns. However, as parameters shrank, results became more useful. The results of each run are as followed: For the first run of apriori, where we are looking for patterns based on region size, these are the main patterns that are being discovered:

- Direct payments from the Social Security Administration dominate the most common sets for every county ratio, excluding size 750,000 - 950,000. The administration appears commonly with size 0 - 100,000 with support value of 0.091, size 100,000 - 200,000 with support value of 0.098, size 200,000 - 750,000 with support value of 0.084, and size >950,000 with support value of 0.05. While these support values may appear small, they reveal large amounts of social security funding in all of these regions, ranging from 50 - 150 transactions each.
- Regions of size 750,000 - 950,000 appear the most in receiving delivery orders from the Department of Defense, with support of 0.039. This is due to the presence of Lockheed Martin Corp in Cobb County, which is a top recipient of contracts, and therefore skews the award type frequency in this area. The only other non-social security awards that frequently appear in this region are awards made to Emory University in DeKalb, with delivery order contracts appearing at a support value of 0.021 and cooperative agreements appearing at support value of 0.016.
- Only larger regions appear to receive different grant types from a larger variety of agencies. Of the sets with relevant support, Fulton is the only county to receive block grants (Department of Health and Human services with support value 0.023) or formula grants (Department of Education at support value of 0.016, Department of Transportation at support value 0.035, and Department of Agriculture at support value 0.011).
- The Department of Veterans Affairs appears commonly in all regions, however unlike social security, they appear more frequently in smaller regions (Size 100,000 - 200,000 at support value 0.022 and 200,000 - 750,000 at support value 0.029, specifically in regions closer to size 250,000).

Hundreds of association rules were found that helped support the validity of the findings. Larger counties had a more variety of awarding agencies and funding types compared to more rural areas, with rules displaying confidence that ranged from 0.8 - 1, with lift values between 3 and 4. This is expected, as larger counties with large cities tend to house the majority of larger businesses and government organizations. Smaller counties had almost exclusive frequency with the Social Security Administration and the Department of Veterans Affairs. Each rule that displayed one of these two agencies given one of the smaller counties had a confidence value of 1, and a lift value averaging 6. This reveals the sheer amount of individuals in Georgia that receive government benefits related

to retirement programs, and furthermore reveals the number of veterans in smaller areas that benefit from government support, which is a surprising metric compared to urban areas.

IV. RESULT (AGENCY - BUSINESS TYPE - COUNTY RELATIONS)

When the parameters were shrunk to awarding agency, business type, and county, there were some surprising patterns, but the data mostly confirmed what we found in the first run. Larger counties, such as Fulton, appear in a bigger variety of rules than smaller counties due to the higher concentration of businesses found within them, while smaller counties appear in less rules and exclusively appear at high support values given that the funding agency is the Social Security Administration. Both the Department of Defense and Corporate companies within Fulton and Cobb exclusively co-occur at support values ranging from 0.03 with a confidence of 0.76. less frequent departments, such as the Department of Transportation and the Department of Agriculture, all exclusively support State Governments found in Fulton County. These results only demonstrate the specialized nature of each federal department in who they fund, which is a trivial metric like the results of the original run.

There are two surprising results found. First, the Department of Veterans Affairs appears with INDIVIDUAL at a high support of 0.07, and a high confidence of 0.95, further reinforcing the high appearance of this department in various low population counties. This can be due to a high concentration of veterans in smaller counties or a low support of veterans in higher counties. This is unlikely to be ill-gotten funding due to the previous discovered variety of veterans assistance in various counties. Therefore, this could be an indication of the rising need for veterans support across the state. Second, the Department of Health and Human services frequently co-occurs in multiple business types, those being NONPROFIT WITH 501C3 IRS STATUS (support of 0.03 and confidence of 0.8), Corporate Non-Tax Exempt (support of 0.02 and confidence of 0.74), and Private Education Institutions (support of 0.01 and confidence of 1). Since there is a Georgia Department of Public Health and a Georgia Department of Human Service, it is surprising to not see frequent co-occurrences between them and the federal department. Furthermore, it is surprising to see frequent co-occurrences with DeKalb, and not Fulton County, however DeKalb is home to many large recipients of federal funding, such as Emory University, which most likely contributes to this discovery. Even so, it is surprising to see these companies receive more funding than the state government.

V. RESULT (TRANSACTION DESCRIPTION: COMMON WORD SETS)

In the third run, the descriptions of the transaction purposes did not reveal any surprising information we did not see in the original run, but rather revealed confirmation of what we saw before. There is only one set of meaningful words that frequently occur, those being "Benefits", "Aggregate",

”(RI)”, ”Retirement”, and ”Insurance”, which all co-occur with each other at support value of 0.37. These are all words that are used to describe transactions from the Social Security Administration, which reinforces their established rank of highest funding agency. This could also reveal the lack of any common interests among the awarding agencies and a nature of specific goals for each transaction, however this finding only shows a lack of patterns that we are looking for.

VI. RESULT (AGENCY - RECIPIENT RELATIONS)

We began to see more surprising results in the agency-recipient relations. There appears to be a sizable difference between recipients funded by one agency vs multiple agencies, with there being more recipients with exclusive relations with a specific agency. This is not surprising on its own, however it also appears that the recipients who receive more than one transaction are the ones with exclusive relations. There are a hand full of recipients who have received transactions from multiple agencies, all of which are once of three kinds of businesses: a Georgia Agency, large university, or miscellaneous business who has received one transaction at some point from one of the leading agencies in transactions, those being the Department of Defense, Department of Health and Human Services, and Department of Education. Recipients who have transactions with multiple agencies due to one transaction don’t have a high amount of collective transactions and become pruned when raising the minimum support value. The best way to interpret this is that the recipients with lower transaction counts are those who request funding from a specific agency for a specific purpose or project, while those who receive multiple transactions are mostly businesses who might operate off of the funds of one specific agency who funds them. While this may not be surprising, it is interesting to see the way the results shape this rule, and it provides more interesting and surprising rules than the runs that use multiple parameters.

VII. RESULT (K-MEANS)

Similar to the apriori runs, we used k-means to limit the amount of variables to the recipients, and it began to provide surprising and useful results. We performed the elbow method to determine the initial amount of clusters, which revealed 3, however through multiple iterations, we divided the recipients further into five clusters. These clusters can be categorized by these descriptions:

- Group 0: Made up of seven recipients, most of which are state level departments. This category is defined due to the large number of transactions made to these departments, which makes sense since the state departments further distribute the funds to other local recipients. This group receives a large number of transactions and dollars compared to the rest of the recipients.
- Group 1: Represented by ”Multiple Recipients”, which makes up those protected individuals funded by social programs from entities like the Social Security Administration. These individuals collectively make up the

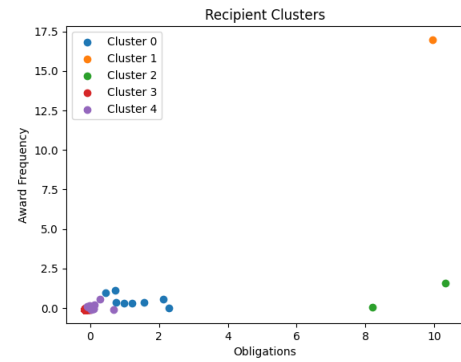


Fig. 4. Graph displaying every recipient and which cluster they belong to.

recipient with the largest number of awards, and one of the largest amounts of money.

- Group 2: Houses Lockheed Martin Corp and the Georgia Department of Community Health, both of which are the other two highest recipients of federal funds. They both received less than 100 contracts over the original threshold of 20,000,000, and therefore do not join group 1 with the multiple recipients of social funds, but still receive enough to create their own category. What is surprising is that Lockheed Martin receives more federal funds than the individuals on social programs. This is an interesting metric that could be interpreted many ways, however without comparing this metric to other fiscal years, it is hard to tell if this is coincidence or not.
- Group 3: The smaller businesses who have received less than 10 contracts, and do not receive as much funding as the rest of the recipients. Most of these recipients have only received 1 transaction, and were pruned from the list of frequent sets done by association. These are the recipients who most likely do not operate mostly on federal funds, however have received grants for specific projects. This category also houses some universities and state departments who have received a minimal amount of contracts compared to their like recipients, and therefore did not make the category with them.
- Group 4: A miscellaneous category. These are recipients who have either received a large amount of funding compared to the low amount of transactions they have received, such as Peraton Inc., or have a low amount of federal funding compared to the high amount of contracts they have received, such as Emory University. There are no real connecting features that define these recipients together, however they appear to be either state departments or universities who didn’t receive enough to be in Group 0, or businesses who received too much to be in Group 3.

VIII. INTERPRETATION

First we asked ”What combinations of awarding agencies, recipients, and award types frequently occur in different sized regions in Georgia?” We found that larger regions generally

hold the largest state government programs, military industries, and universities that receive the majority of federal funding. Furthermore, we found that the majority of frequent recipients across the board only co-occur exclusively with one agency, and those who co-occur with multiple agencies do not receive many transactions. Smaller regions are less likely to receive funding from anywhere other than social security. While some of these findings are surprising to see visualized, they are not out of the ordinary.

Apriori demonstrated more surprising results when the frequent sets were made up of a smaller amount of variables. This means that it is more effective to examine one parameter based on other variables rather than examining combinations of several variables. With the sheer quantity of transactions, and the sheer variety in transaction details, examining one of those details alone shows more focus and reveals a lot more than the broader scope. This is confirmed once more in the results of k-means, which examined only the characteristics that define the recipient in the given transaction. While the results were not out of the ordinary, it is surprising to see the natural clustering of these recipients and how they compare to each other rather than their individual relations with other qualitative variables.

Second, we asked "What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?" The largest found group of recipients is the group of small businesses who receive less than 10 transactions, who most likely receive sparing transactions for specific purposes. On the other side of the spectrum is the group of state facilities who receive the majority of funding and receive a frequent amount of transactions. Beyond this is two groups, one held exclusively by individuals who receive social programs, and one held by Lockheed Martin and the Georgia Department of Community Health, who are also in the top three recipients in transactions and funding, however they do not receive as many transactions as the collective of individuals on social programs. Lastly, there is a miscellaneous category for larger firms who did not receive enough to be in the state firm category, or smaller businesses who received too much to be in their respective category.

Lastly, there does not appear to be any real characteristics that define anomalously high transactions. These transactions are mostly defined either by their high value, or by their high amount of COVID or IJIA disaster reliefs. However, there are no qualitative characteristics these transactions have in common, or that are out of the ordinary for federal funding. All of these results suggest that, at the highest level of funding, Georgia tends to follow typical federal funding trends with no real red flags to investigate.

IX. VALIDITY

We are confident that the results we have received are valid and hold up to the real world expectations. All of the results generated from each data mining technique mostly validate trivial geographic patterns and expected patterns of money flow. In the four runs of apriori we performed, patterns

revealing high social program funding and large variety in transaction types in large, urban areas proved consistent in high confidence values and high lift values, increasing the validity of the results. K-means was ran under 10 random seeds, and was also ran with three clusters in addition to five clusters, and the resulting groups proved similar every time. The main difference between having three clusters and five is the extra separation of the largest cluster, which stayed consistent in both runs. Some preprocessing choices could have had major effects on the results. Primarily, many of the transactions had addresses listed for out of state locations, but were clearly for Georgia locations. This was solved by finding the closest matching zip code for each of these businesses. While this prevented the elimination of a large amount of the data, it could have heavily skewed the results of the data, however we are still confident that the results are valid because we examined the top percentage of transactions, a window that should reflect the normal funding patterns of Georgia.

X. LIMITATIONS

While the rules generated by the apriori algorithm are relevant with strong support, the sheer volume of transactions and different recipients makes it difficult to find patterns that are not obvious. The descriptions of each transaction vary heavily, making it nearly impossible to use them as a factor in finding frequent sets of transaction characteristics among different regions. Furthermore, without examining other states in the same manner, it is difficult to display the weight of the significant rules found here. In the future, this same algorithm should be performed in other states of similar size in order to compare their sets. This would better reveal of the high number of social security transactions is exclusive to Georgia, or if they are the largest distribution of federal funds in the country.

Another limitation that may have effected the results is the categories used to create some of the sets. The county names were kept in addition to their sizes in order to help explain common sets further. However, with the imbalance in counties per size ratio, some sets appear as a consequence of county characteristics being paired with their names. Furthermore, certain recipients only receive specific types of funds from specific agencies. While the less frequent recipients were naturally filtered out, this lead to frequent sets appearing with identical information, but different support values. This was mitigated by favoring certain category combinations over others, such as awarding agencies paired with county sizes, however it is possible that this could have skewed the results and led to the large amount of trivial patterns.

Furthermore, as we performed more mining, the limitations of our prerequisites appeared more. We chose to only examine transactions with a minimum obligation of 20,000,000, a decision to reduce dimensionality and make the data more manageable in the given time and resources. This may help in those regards, however there is a chance that this limitation has skewed the data in major ways. For example, there is a significant chance that some of the smaller businesses receive

more transactions than is present in the current data set that are not being accounted for. In addition to this, we mentioned that Lockheed Martin received more total obligations than the collective of individuals on social programs. This could be a real possibility given the state of current events in regards to military action, however there is no way of telling if this is a legitimate metric without the total picture of transactions.

We also chose to limit our scope to fiscal year 2025, a decision made for reducing dimensionality, and also to limit the amount of skewed data that may come from any abnormal fiscal years. However, this only increases the amount of trivial data, for without a direct comparison to other fiscal years, it is hard to validate the results of our study. Furthermore, without comparing the results to another state, we cannot tell how surprising our results are. However, since the results of this study are surprising, but not out of the ordinary, they can be used as a direct comparison to other states or other fiscal years to judge the surprising nature of other data sets.

XI. ETHICS

United States federal spending is a completely ethical area of examination because these government records and programs are public knowledge to the wide public under the Freedom of Information Act. The website used to retrieve the dataset is a public .gov site updated by the U.S. Department of the Treasury, where every record of federal spending to state level affairs is available to view and use. Most businesses that receive government funding make this fact known, and these records are available on multiple sites. Privacy is of no concern either, because we do not use any information from any specific individual, as individuals who receive social security or similar funds are labeled under "MULTIPLE RECIPIENTS" to keep their identities anonymous. Overall, this information is protected under the Freedom of Information Act and is free for anybody to use or examine.

XII. CONCLUSION

Georgia appears to follow typical federal spending patterns and geographical rules. Larger counties appear to frequently co-occur with more funding agencies and receive the most variety of transaction types, while smaller regions appear to almost exclusively receive social program funding. Five main groups of recipients appear in Georgia that receive federal funding: recipients of social security, the top recipients of federal dollars (Lockheed Martin and the Georgia Department of Community Health), state departments, small businesses, and miscellaneous businesses who receive disproportionate amounts of awards or obligations. Finally, there do not appear to be any abnormal characteristics that categorize anomalous transactions, with COVID prevention and reimbursement being the main defining feature.

Since Georgia follows typical spending patterns, others who are looking at the federal patterns of their own states would do well using Georgia as a direct comparison to check for any abnormal patterns. For the future of this work, we want to perform this comparison to other states to further validate

our findings of Georgia. Furthermore, we want to see if our findings hold up when we expand the minimum transaction value and compare the results to other fiscal years. From the typical results of Georgia funding, it should provide confidence to its residents that their tax dollars are not being wasted in mass on any fraudulent or unnecessary expenditures, but rather are being used responsibly for the benefit of the state.

REFERENCES

- [1] "State Profile: Georgia." *USAspending.gov*, <https://www.usaspending.gov/state/Georgia>. Accessed January 15, 2026.